

Replace a Filter Cartridge SOP

This SOP covers replacing the cartridge in a single inline filter housing using a two-arm robot system. Both arms are used throughout: the grippers cooperate to hold the housing, unscrew the sump, swap the cartridge, seat the O-ring, rescrew the sump, wipe the housing down and apply a date label, with the left and right grippers taking the specific roles called out in each step. The task runs from a housing with a spent cartridge in the start zone to a serviced, labelled housing in the output zone.

The table is set up in one of three ways. Only the housing and the new cartridge move; the cradle, the tool rest and the waste bin are in the same place in all three.

- **Config L:** the housing and the new cartridge are at the back-left. Output at the back-right.
- **Config M:** the housing and the new cartridge are at the front-centre, in front of the cradle. Output at the back-right.
- **Config R:** the housing and the new cartridge are at the back-right. The output zone moves to the back-left.

Where a step depends on the setup it says so on an **IF** line — look at the table and follow the line that matches.

What stays constant across all sessions:

- **Start position:** the housing and the new cartridge start at the back-left (**Config L**), the front-centre (**Config M**) or the back-right (**Config R**). One config per episode, chosen before recording and never changed mid-episode.
- **Same-side rule:** the gripper on the start zone's side picks the housing and the new cartridge — the left gripper in Config L and M, the right gripper in Config R. No arm reaches across the table.
- **Hand-over rule:** in Config L and M the new cartridge is on the left gripper's side but the right gripper fits it, so the left gripper **hands it over** above the service station: the left gripper holds the cartridge still, the right gripper closes on the opposite side of it, and only then does the left gripper open and lift clear. Nothing is handed over in Config R.
- **Fixed roles:** in every config the left gripper holds the cap in the cradle and the right gripper works the wrench, the tools, the cartridge and the O-ring. The cradle, the tool rest and the waste bin never move; only the output zone swaps to the back-left in Config R.
- **Working position:** the housing is always moved to the service cradle at the centre of the table before the sump is unscrewed or any cartridge is handled.
- **Orientation target:** every new cartridge must go in **gasket end up, flow arrow pointing toward the cap, and centred on the standpipe**, before the sump is rescrewed.
- **Cycle order:** unscrew the sump, swap the cartridge, seat the O-ring, rescrew the sump, wipe the housing, then date-label it. One cartridge per housing, one housing per episode.

- **Pass condition:** a housing is only staged as finished once the sump is snug, the O-ring is seated in its groove, the exterior is dry and a legible date label is applied to the cap.

Setup

Go through both checklists before starting the episode.

Hardware checklist

- Cameras are on and recording
- Env camera frame includes the front and back edges of the table and is centered on the table's midpoint
- Both arms are at the home position with grippers open
- Table surface is clear of any objects other than the cradle, the tools, the waste bin and the housings and cartridges

Materials checklist

- Filter housing to be serviced is placed in the start zone for this episode's config, cap up, with a spent cartridge inside and the sump hand-tight
 - **Config L:** back-left of the table
 - **Config M:** front-centre of the table, in front of the cradle
 - **Config R:** back-right of the table
- New cartridge is placed in the same start zone beside the housing, in its sleeve, gasket end up
- Service cradle is at the **center** of the table, empty and dry
- Housing wrench, lubricant, wipe cloth, label roll and marker are on the tool rest at the **front-right**
- Waste bin is at the **front-left** of the table, open and empty
- The output location for the serviced housing is clear: back-right (Config L and M) or back-left (Config R)

Workspace layout

- **Start zone:** housing to be serviced and new cartridge (input) — back-left (Config L), front-centre (Config M) or back-right (Config R)
- **Center:** service cradle (unscrew + swap + seat + rescrew + wipe + label)
- **Front-left:** waste bin for the spent cartridge
- **Front-right:** tool rest (wrench, lubricant, cloth, label roll and marker)
- **Back-right:** serviced housing (output) in Config L and M; **back-left** in Config R

Vocabulary

These are the terms used in this SOP. Operators and annotators must use this language consistently. One term per concept, used throughout.

Housing anatomy

- **Filter housing:** the two-piece canister that holds one cartridge; a cap and a sump.
- **Cap:** the fixed upper half of the housing carrying the inlet and outlet ports.
- **Sump:** the removable lower bowl of the housing that screws onto the cap.
- **Sump threads:** the coarse threads at the top of the sump that mate with the cap.
- **O-ring:** the rubber sealing ring that sits between the cap and the sump.
- **O-ring groove:** the machined channel at the top of the sump that the O-ring sits in.
- **Standpipe:** the post inside the cap that the cartridge slides down over and centres on.
- **Sump wrench flats:** the two flat faces on the outside of the sump the wrench grips.

Cartridge anatomy

- **Cartridge:** the replaceable filter element inside the housing.
- **Spent cartridge:** the used cartridge removed from the housing; it goes to the waste bin.
- **New cartridge:** the unused replacement cartridge taken from the input zone.
- **Gasket end:** the end of the cartridge carrying the moulded sealing gasket; the gasket end always goes up, against the cap.
- **Open end:** the end of the cartridge opposite the gasket end; it rests in the bottom of the sump.
- **Flow arrow:** the printed arrow on the cartridge label showing the direction of flow.
- **Seated:** the cartridge has stopped moving down the standpipe and is fully home in the sump.
- **Cocked:** the cartridge is tilted, so its axis is not parallel to the sump axis.
- **Proud:** the cartridge stands above the O-ring groove, so the sump cannot close on the cap.
- **Snug:** the sump is turned until it stops on the O-ring, plus about a quarter turn by wrench.

Tooling

- **Service cradle:** the moulded rest at the center of the table that holds the cap while the sump is off.
- **Housing wrench:** the band or strap wrench used to break the sump loose and to snug it back up.
- **Lubricant:** the tub of food-grade silicone grease used on the O-ring.
- **Wipe cloth:** the lint-free cloth used to dry the housing exterior and the O-ring groove.
- **Label roll:** the roll of blank date labels on the tool rest.

- **Marker:** the pen used to write the service date on the label.
- **Tool rest:** the marked spot at the front-right of the table where the tools live.
- **Waste bin:** the open bin at the front-left of the table that the spent cartridge goes into.

Workspace zones

- **Start zone:** input zone holding the housing to be serviced and the new cartridge — back-left (**Config L**), front-centre (**Config M**) or back-right (**Config R**). One per episode, chosen before recording and never changed mid-episode.
- **Service station:** the center of the table where unscrewing, swapping, rescrewing, wiping and labelling happen.
- **Output side:** where the serviced housing is placed — the back-right of the table in Config L and M, the back-left in Config R.
- **Home position:** the default resting pose for each arm: gripper open and clear of the table.

Actions

- **Pick:** move one housing, or one cartridge, to the service station.
- **Unscrew:** turn the sump anti-clockwise off the cap, breaking it loose with the wrench first.
- **Swap:** lift the spent cartridge out, discard it, and lower the new cartridge in gasket end up.
- **Seat:** lay the lubricated O-ring flat in its groove all the way round.
- **Rescrew:** turn the sump clockwise back onto the cap by hand, then snug it with the wrench.
- **Wipe:** dry the housing exterior and the cap with the cloth.
- **Date-label:** write the service date on a label and press it onto the cap.
- **Stage:** place the finished housing in the output zone (batch sessions place them in a row).
- **Hand over:** the left gripper holds the new cartridge still above the service station, the right gripper closes on the opposite side of it, and only then does the left gripper open and lift clear. Config L and M only, because the right gripper fits the cartridge.
- **Next housing:** return to Step 1 with the next housing until the session target is done.

Steps

Only Step 1, Step 4.1 and Step 9 depend on the config: the **start-zone-side gripper** picks the housing and the new cartridge (left in Config L and M, right in Config R), in Config L and M the new cartridge is handed over to the right gripper above the service station, and in Config R the serviced housing is staged at the back-left instead of the back-right. Every other line is the same in all three configs.

Step 1: Move a housing to the working area

Goal: one filter housing sits upright in the service cradle, ready for the sump to come off.

Look where the housing is before reaching for it.

- **IF the housing is at the back-left (Config L):** with the **left gripper**, grasp the housing by the cap and carry it forward and to the right to the service cradle at the center of the table.
- **IF the housing is at the front-centre (Config M):** with the **left gripper**, grasp the housing by the cap and carry it straight back to the service cradle at the center of the table.
- **IF the housing is at the back-right (Config R):** with the **right gripper**, grasp the housing by the cap and carry it forward and to the left to the service cradle at the center of the table.

Then, in all three:

- Set it down in the cradle cap up.
- Press it down into the cradle until it sits fully seated with no rock, then release.

Step 2: Unscrew the sump from the cap

Goal: the sump is off the cap and set down open end up, with the cap still seated in the cradle and the spent cartridge exposed.

2.1 Hold the cap in the cradle

- With the left gripper, hold the cap down against the cradle so the housing cannot turn or lift.
- Keep this hold for the whole of Step 2; the cap never turns, only the sump does.

2.2 Fetch the housing wrench

- With the right gripper, grasp the housing wrench from the tool rest at the front-right of the table.
- Bring it to the service station and fit it onto the sump wrench flats, square to the sump.

2.3 Break the sump loose

- Turn the wrench **anti-clockwise** in one steady pull until the sump breaks loose from the cap.
- If the sump does not move on the first pull → re-seat the wrench square on the flats and pull again. Do not lever the wrench off-axis or strike it.

2.4 Return the wrench and spin the sump off

- Lift the wrench clear and place it back on the tool rest at the front-right, then release it.
- With the right gripper, grasp the sump body and turn it anti-clockwise by hand until it comes free of the cap threads.
- Lower the sump straight down so the cartridge is not dragged sideways against the standpipe.

2.5 Set the sump down

- Set the sump on the table beside the cradle, open end up, so it cannot roll.
- Release the sump with the right gripper and release the cap with the left gripper.

Step 3: Remove the spent cartridge

Goal: the spent cartridge is out of the housing and in the waste bin at the front-left.

3.1 Lift the spent cartridge out

- With the right gripper, grasp the spent cartridge by its outside diameter.
- Lift it straight up and out, clear of the standpipe and the O-ring groove, without knocking the groove.

3.2 Discard it

- Carry the spent cartridge to the waste bin at the front-left of the table and release it into the bin.
- Do not set the spent cartridge back on the table and do not return it to the input zone.

Step 4: Fit the new cartridge with the correct orientation

Goal: the new cartridge stands in the housing gasket end up, flow arrow toward the cap, square to the sump axis and centred on the standpipe.

4.1 Find and pick up a new cartridge

- **IF Config L or M:** with the **left gripper**, grasp one new cartridge from the start zone by its outside diameter. **IF Config R:** with the **right gripper**, grasp one new cartridge from the back-right by its outside diameter. If no cartridge is clear of the others, rotate/reorient anti-clockwise until one comes free, then grasp it.
- Raise it clear of the input pile and carry it to the service station. **IF Config L or M:** hand it over above the service station — the left gripper holds the cartridge still, the **right gripper** closes on the opposite side of it, then the left gripper opens and lifts clear. The right gripper holds the cartridge from 4.2 on in all three.

4.2 Check orientation - gasket end up

- If the gasket end points up, toward the cap, and the flow arrow points the same way → continue to 4.3.
- If the gasket end points down → hand the cartridge to the left gripper, regrip it from the opposite end with the right gripper, and turn it 180° so the gasket end and the flow arrow face the cap.

4.3 Lower the cartridge in

- Lower the cartridge straight down into the sump until the open end rests on the bottom of the sump.
- Release slowly. Do not force the cartridge down or twist it into place against the sump wall.

4.4 Check alignment - square and centred

- If the cartridge stands square, with its gasket end level and clear of the O-ring groove → continue to 4.5.
- If the cartridge is cocked → with the left gripper, grasp it by the outside diameter, lift it clear, reset it straight down and release. If it cocks again → return to 4.3.

4.5 Check the cartridge is not proud

- Confirm the gasket end sits below the top of the sump, so the O-ring groove is fully clear and the sump can close on the cap.
- If the cartridge stands above the groove → lift it out and confirm it is the right cartridge for this housing before refitting.

Step 5: Seat the O-ring

Goal: a clean, lightly greased O-ring lies flat in its groove all the way round, with no twist and no roll-out.

5.1 Lift the O-ring out and inspect it

- With the right gripper, lift the O-ring out of its groove at the top of the sump.
- Check it for cuts, flats, splits or grit. If it is damaged → discard it in the waste bin and take a replacement O-ring from the tool rest.

5.2 Wipe the groove

- With the right gripper, grasp the wipe cloth from the tool rest and wipe the O-ring groove clean and dry.
- Return the cloth to the tool rest. Grit left in the groove is the most common cause of a leak.

5.3 Grease the O-ring

- With the left gripper, hold the O-ring; with the right gripper, take a small amount of lubricant from the tub.
- Spread a thin, even film over the whole outside of the O-ring. Do not fill the groove with grease.

5.4 Seat it in the groove

- Lay the O-ring into the groove and press it down all the way round until it lies flat and even.
- Confirm the O-ring is not twisted, is not pinched over the groove edge, and does not stand above the sump face at any point, then release.

Step 6: Rescrew the sump onto the cap

Goal: the sump is back on the cap, square, hand-tight and then snug with the wrench, with the O-ring still in its groove.

6.1 Offer the sump up square

- With the left gripper, hold the cap down against the cradle.
- With the right gripper, raise the sump straight up under the cap so the standpipe enters the cartridge and the threads meet square, with no cross-threading.

6.2 Hand-tighten

- Turn the sump **clockwise** by hand in one continuous motion until it stops firmly against the O-ring.
- If the sump binds or turns hard early → back it off anti-clockwise all the way, check the cartridge and the O-ring, and return to 6.1.

6.3 Snug with the wrench

- With the right gripper, fetch the housing wrench from the tool rest and fit it square on the sump wrench flats.
- Turn it clockwise about a quarter turn past hand-tight, then stop. Do not keep pulling past snug.
- Lift the wrench clear, return it to the tool rest at the front-right, and release it.

Warning: The housing should now be closed with the new cartridge inside, the O-ring seated and the sump snug, ready to be wiped and labelled.

Step 7: Wipe the housing

Goal: the outside of the housing is dry and clean, with no grease, water or fingerprints left on it.

- With the right gripper, grasp the wipe cloth from the tool rest at the front-right of the table.
- Wipe the sump from the top down and around, then wipe the cap, including the area the label will go on.
- With the left gripper, hold the housing in the cradle so it does not turn or lift while it is wiped.
- Return the cloth to the tool rest and release it. The cloth never stays on the housing or in the cradle.

Step 8: Apply the date label

Goal: one legible date label is pressed flat onto the cap of the serviced housing.

- With the right gripper, take one label from the roll on the tool rest and the marker beside it.
- Write the service date on the label, then set the marker back on the tool rest.
- Press the label flat onto the wiped area of the cap, right side up, with no wrinkle and no overhang past the cap edge.
- Confirm the date reads clearly in the env camera frame, then release.

Step 9: Move the serviced housing to the output zone

Goal: the finished housing rests upright in the designated output zone.

- **IF Config L or M:** with the **right gripper**, grasp the serviced housing by the cap and move it back and to the right into the output location at the back-right of the table. **IF Config R:** with the **left gripper**, grasp the serviced housing by the cap and move it back and to the left into the output location at the back-left of the table.
- Set it down cap up.
- Place it alongside any housings already there, caps all up and labels facing front, within 10-20 cm and not touching.
- Release the gripper.

Step 10: Return to home and end the episode

- Move both arms back to the home position with grippers open.
- End data collection.

SOP violations

These are the things the operator can do that break (violate) the SOP. This list feeds the violation set and is what reviewers look for using the side-by-side review tool.

How to record a violation in review

For each violation you spot in a recorded episode, record:

- The **start timestamp** of the violation in the video.
- The **violation name** from the list below.
- The **SOP rule broken** (the step number from the list below).

The **visible cue** is what you actually see in the video. The **coaching note** is for retraining the operator after the review; it is not what the annotator labels.

Episode handling

Any SOP violation is flagged (or tagged) with the timestamp and violation name. Rather than being discarded, the episode is retained in the training data and tagged with the violation. No flagged episode is deleted. This matches the goal of capturing realistic operator variance in training.

Violations

Note on the start position: the violations below were written for Config L (housing and new cartridge at the back-left, output at the back-right). The pickup and arm-role cues will be

rewritten later to cover all three start positions; they are left as they are for now. Until then, anything that does not match the episode's config goes under **Config misaligned**.

Violation: Config misaligned.

- **Visible cue:** what the operator does does not match the config on the table — the housing or the new cartridge is not in the start zone for the config; a gripper reaches across the table for the housing or the cartridge; in Config L or M the left gripper lowers the cartridge into the sump itself, or the right gripper takes it from the start zone, instead of a hand-over above the service station; the serviced housing is staged on the wrong side; or the wrong IF line is followed.
- **SOP rule broken:** the start position, the same-side rule and the hand-over rule (the left gripper picks in Config L and M and hands the cartridge to the right gripper; the right gripper picks in Config R and the housing is staged at the back-left; the IF line followed is the one for the config on the table).
- **Coaching note:** look where the housing is before the first reach, then follow that config's IF lines through Steps 1, 4.1 and 9.

Violation: Wrong pickup.

- **Visible cue:** a housing or a cartridge is started from the center or right instead of the back-left input zone, it is placed outside the service cradle, or it is dragged or slid across the table instead of being picked up (lifted clear of the surface) and carried.
- **SOP rule broken:** Steps 1 and 4.1 (pick from the back-left input zone and bring it to the service station). Pick means lift and carry, not drag.
- **Coaching note:** always start the housing and each cartridge from the back-left and bring it to the center before opening anything. Lift it clear of the table and carry it. Do not drag or slide it across the surface.

Violation: Picked more than one housing or cartridge at once.

- **Visible cue:** two or more housings, or two or more new cartridges, are lifted/moved from the input zone in a single pick.
- **SOP rule broken:** Steps 1 and 4.1 (move one housing, and one cartridge, per cycle).
- **Coaching note:** pick exactly one housing and one new cartridge per service cycle.

Violation: Housing not seated in the cradle before the sump is turned.

- **Visible cue:** the wrench is fitted or the sump is turned while the housing rocks in the cradle, sits on the table beside it, or is held only in the air.
- **SOP rule broken:** Steps 1 and 2.1 (seat the housing fully in the cradle and hold the cap down before turning the sump).
- **Coaching note:** seat the housing in the cradle and hold the cap before any turning force goes on the sump. A loose housing turns with the sump and cross-threads the cap.

Violation: Cap turned instead of the sump.

- **Visible cue:** the cap is rotated in the cradle while the sump is held still, or both are turned together, during Step 2 or Step 6.
- **SOP rule broken:** Step 2.1 (the cap never turns, only the sump does).
- **Coaching note:** hold the cap still and turn the sump. Turning the cap twists the plumbing side of the housing.

Violation: Wrong turn direction on the sump.

- **Visible cue:** the sump is turned clockwise to open it in Step 2, or anti-clockwise to close it in Step 6.
- **SOP rule broken:** Steps 2.3 and 6.2 (anti-clockwise to open, clockwise to close).
- **Coaching note:** anti-clockwise off, clockwise on. Forcing the wrong direction rounds the threads.

Violation: Wrench used incorrectly, or not used to break the sump loose.

- **Visible cue:** the sump is broken loose by hand force alone, the wrench is fitted off the wrench flats or off-square, it is levered off-axis, or it is struck to free the sump.
- **SOP rule broken:** Steps 2.2 and 2.3 (fit the wrench square on the flats and break the sump loose in one steady pull).
- **Coaching note:** fit the wrench square on the flats and pull steadily. Hand force alone slips, and levering or striking the wrench damages the sump.

Violation: Wrench or cloth not returned to the tool rest.

- **Visible cue:** the wrench, the cloth, the lubricant tub or the marker is left on the housing, in the cradle, on the table, or anywhere other than the tool rest at the front-right.
- **SOP rule broken:** Steps 2.4, 5.2, 6.3, 7 and 8 (return each tool to the tool rest and release it).
- **Coaching note:** every tool goes back on the rest as soon as it is done with, so the station stays clear for the next step.

Violation: Sump dragged sideways off the cap.

- **Visible cue:** the sump is pulled off at an angle or swung sideways, dragging the cartridge across the standpipe or the O-ring groove, instead of being lowered straight down.
- **SOP rule broken:** Step 2.4 (lower the sump straight down off the threads).
- **Coaching note:** come straight down off the cap. A sideways pull scores the groove and the standpipe.

Violation: Sump left where it can roll or be knocked over.

- **Visible cue:** the sump is set down on its side, on the cradle, on the tool rest, or open end down, or it is left in the arm's path.
- **SOP rule broken:** Step 2.5 (set the sump beside the cradle, open end up, so it cannot roll).
- **Coaching note:** park the sump open end up beside the cradle every time so nothing spills and nothing rolls.

Violation: Spent cartridge not discarded.

- **Visible cue:** the spent cartridge is set back on the table, left in the sump, put on the tool rest, or returned to the back-left input zone instead of going into the waste bin.
- **SOP rule broken:** Step 3.2 (carry the spent cartridge to the waste bin at the front-left and release it there).
- **Coaching note:** the spent cartridge goes straight into the bin. A used cartridge left on the table gets refitted by mistake.

Violation: Cartridge installed backwards (gasket end down).

- **Visible cue:** the new cartridge is lowered in with the gasket end down and the open end up, or with the flow arrow pointing away from the cap.
- **SOP rule broken:** Orientation target and Step 4.2 (gasket end up, flow arrow toward the cap).
- **Coaching note:** the gasket end always goes up against the cap. Check the end and the flow arrow before releasing the cartridge.

Violation: Cartridge not correctly found before grasping.

- **Visible cue:** the operator fails to find and grasp a new cartridge: the grip closes on empty table, on the housing, or on two cartridges at once; or, when no cartridge is clear of the others, the operator reorients clockwise (or in an arbitrary back-and-forth direction) instead of anti-clockwise to free one.
- **SOP rule broken:** Step 4.1 (grasp one cartridge by its outside diameter; if none is clear, rotate/reorient anti-clockwise until one comes free, then grasp it).
- **Coaching note:** grasp one cartridge on its outside diameter and confirm it is actually in view and clear before closing the gripper. If none is free, reorient anti-clockwise (a consistent search direction) until one comes loose, then grasp it.

Violation: Cartridge left cocked or proud.

- **Visible cue:** the sump is offered back up while the cartridge is tilted in the sump, is resting on the O-ring groove, or stands above the top of the sump.
- **SOP rule broken:** Steps 4.4 and 4.5 (square to the sump axis, seated, and clear of the O-ring groove).

- **Coaching note:** settle the cartridge square and fully down before the sump goes back on. A cocked or proud cartridge is what crushes the O-ring.

Violation: Cartridge forced or twisted into the sump.

- **Visible cue:** the cartridge is pushed down hard, screwed or twisted against the sump wall, or driven in with gripper force rather than lowered straight down and released.
- **SOP rule broken:** Step 4.3 (lower it straight down until the open end rests on the bottom of the sump, then release slowly).
- **Coaching note:** the cartridge drops in under its own weight. Forcing it deforms the gasket end.

Violation: O-ring not lifted out or not inspected.

- **Visible cue:** the O-ring is left in the groove for the whole cycle, or it is refitted without being looked over for cuts, flats or grit.
- **SOP rule broken:** Step 5.1 (lift the O-ring out and inspect it).
- **Coaching note:** every cycle the O-ring comes out and gets looked at. A cut or flattened O-ring will leak no matter how snug the sump is.

Violation: O-ring groove not wiped.

- **Visible cue:** the O-ring is put back into a groove that has not been wiped clean and dry with the cloth.
- **SOP rule broken:** Step 5.2 (wipe the O-ring groove clean and dry).
- **Coaching note:** wipe the groove before the O-ring goes back. Grit in the groove is the most common leak cause.

Violation: O-ring not greased, or over-greased.

- **Visible cue:** the O-ring is refitted dry, or so much lubricant is used that grease fills the groove, runs down the sump, or ends up on the cartridge or the cap.
- **SOP rule broken:** Step 5.3 (spread a thin, even film over the outside of the O-ring; do not fill the groove).
- **Coaching note:** a thin film is enough. Dry O-rings roll and pinch; flooded grooves push grease into the water path.

Violation: O-ring not seated correctly.

- **Visible cue:** the O-ring is left twisted, pinched over the groove edge, standing above the sump face, or only partly in the groove when the sump is offered back up.
- **SOP rule broken:** Step 5.4 (lay it flat in the groove all the way round, no twist, no roll-out).
- **Coaching note:** run round the whole groove and confirm the O-ring is flat and even before closing the housing.

Violation: Sump cross-threaded on the way back.

- **Visible cue:** the sump is offered up off-square and turned while the threads are not meshed, so it goes on crooked or binds early, and the operator keeps turning instead of backing it off.
- **SOP rule broken:** Steps 6.1 and 6.2 (offer the sump up square; if it binds, back it off all the way and start again).
- **Coaching note:** meet the threads square and feel the first turn. If it binds, back all the way off; never force a cross-threaded sump.

Violation: Sump left hand-tight only (not snugged).

- **Visible cue:** Step 6.3 is skipped: the wrench is never refitted, so the sump is left at hand-tight when the housing is wiped or staged.
- **SOP rule broken:** Step 6.3 (a quarter turn past hand-tight with the wrench).
- **Coaching note:** hand-tight is not snug. Finish with the quarter turn on the wrench or the housing will weep under pressure.

Violation: Sump over-tightened.

- **Visible cue:** the wrench is pulled well past a quarter turn beyond hand-tight, or the operator keeps pulling after the sump has stopped moving.
- **SOP rule broken:** Step 6.3 (about a quarter turn past hand-tight, then stop).
- **Coaching note:** stop at snug. Over-tightening extrudes the O-ring and makes the next service impossible by hand.

Violation: Housing not wiped.

- **Visible cue:** the housing is labelled or staged with visible grease, water, or fingerprints on the sump or the cap, or Step 7 is skipped entirely.
- **SOP rule broken:** Step 7 (wipe the sump and the cap dry, including the label area).
- **Coaching note:** wipe the housing every cycle. A label will not stick to a greasy cap, and wet housings hide weeps at the next check.

Violation: Date label missing, illegible, or applied badly.

- **Visible cue:** no label is applied, the date is not written before the label goes on, the label is wrinkled, upside down, overhanging the cap edge, stuck to the sump instead of the cap, or unreadable in frame.
- **SOP rule broken:** Step 8 (write the date, press one label flat on the cap, right side up, legible in frame).
- **Coaching note:** write first, then press the label flat on the wiped cap. The date is the whole point of the label; it has to read cleanly on video.

Violation: Wrong cycle order.

- **Visible cue:** the operator fits the new cartridge before the spent one is out, rescrows the sump before the O-ring is seated, labels the housing before it is wiped, or stages it before it is snug.
- **SOP rule broken:** Cycle order (unscrew, swap, seat, rescrow, wipe, label): Step 2 must precede Step 3, Step 3 must precede Step 4, Step 5 must precede Step 6, and Step 7 must precede Step 8.
- **Coaching note:** hold the cycle order every time: unscrew, swap, seat, rescrow, wipe, label. Out-of-order cycles put unsealed housings in the output row.

Violation: Next housing started before the current one is staged.

- **Visible cue:** a second housing or cartridge is brought to the service station while the current housing is still open in the cradle or has not yet been moved to the back-right.
- **SOP rule broken:** Step 9 and the reset/repeat section (do not start the next housing until the current one is closed, labelled and staged).
- **Coaching note:** finish one housing completely, stage it, then fetch the next. Two housings at the station confuse the cartridge pairing.

Violation: Serviced housing placement off zone.

- **Visible cue:** in Step 9 the housing is moved with the left gripper, is left in the cradle instead of being moved to the back-right output zone, is set down on its side, or is released at the back-right but ends up noticeably off from the designated zone, without falling or requiring further correction.
- **SOP rule broken:** Step 9 (grasp the housing by the cap with the right gripper and move it back and to the right into the back-right output zone, cap up).
- **Coaching note:** move the serviced housing to the back-right output zone with the right gripper. Do not leave it in the cradle and aim the release point more precisely; set it down cap up in the corner.

Violation: Wrong batch placement.

- **Visible cue:** in Step 9 the new housing is stood on top of an earlier one, is set down touching it, is more than 10-20 cm away, is laid on its side, or has its label turned away from the front while the others face front.
- **SOP rule broken:** Step 9 (alongside, caps up, labels facing front, within 10-20 cm, not touching).
- **Coaching note:** line the finished housings up cap up with the labels facing front; do not stack them.

Violation: Wrong arm used for an action.

- **Visible cue:** any step that specifies the left gripper or the right gripper is performed with the opposite gripper.
- **SOP rule broken:** any step that specifies a gripper, including Steps 1, 2.1, 2.2, 2.4, 3.1, 4.1, 5.1, 5.3, 6.1, 6.3, 7, 8 and 9.
- **Coaching note:** operator confusion about left vs right roles. Walk through the SOP step by step with the operator.

Violation: Re-grip on a pick or grasp.

- **Visible cue:** the operator closes on the housing, a cartridge, the O-ring, the wrench, the cloth or the marker, finds the grip off, opens, and re-grips before lifting more than two times.
- **SOP rule broken:** Steps 1/2.2/3.1/4.1/5.1 (grasp securely, e.g., the cap, the outside diameter, the wrench flats).
- **Coaching note:** approach angle off. Practice the from-above approach so the first grip catches the right point.

Violation: Repeated fiddling with the cartridge, the O-ring, or the label.

- **Visible cue:** the operator makes many small adjustments (more than about two) to settle the housing in the cradle, the cartridge in the sump, the O-ring in its groove, or the label on the cap, rather than settling it in one or two corrections.
- **SOP rule broken:** the swap/seat/label sub-steps (settle in one or two small adjustments).
- **Coaching note:** grip or approach is off, forcing repeated correction. Tighten the approach so the cartridge, the O-ring and the label land close to target.

Violation: Arms not fully at home position at episode end.

- **Visible cue:** in the final frame, both arms are close to the home position but not exactly at it (grripper not fully open, or arm position visibly off home).
- **SOP rule broken:** Step 10 (return to home position with grippers open).
- **Coaching note:** complete the home motion explicitly before ending recording.

Non-violation failures

These are episode failures that are not the operator's fault and do not go in the violation set. They are recorded as system issues, the episode is discarded/deleted, and the episode does not become a coaching point for the operator.

- **Recording stopped or paused mid-episode:** Cause: software or hardware issue with the recording system.
- **Camera dropped frames or lost feed during the episode:** Cause: camera or capture system issue.
- **Hardware fault on the robot arm:** Cause: gripper malfunction, arm position drift, or motor error during the episode.

- **Housing fault:** Cause: seized, galled or stripped sump threads, a cracked sump, or a cradle that will not hold the housing, through no fault of the operator.
- **Defective cartridge or O-ring:** Cause: a swollen, out-of-round or wrong-size cartridge, a cartridge whose gasket end will not clear the O-ring groove in any orientation, or an O-ring that is split or oversize out of the pack, through no fault of the operator. Replace before the next episode.
- **Consumables out:** Cause: the lubricant tub, the label roll, or the marker is empty or dry, so Step 5 or Step 8 cannot be completed. Restock before the next episode.

After each episode: repeat or reset

- **Batch sessions (e.g., 4x):** the housings are serviced one after another within the single episode. Once each one is snug, wiped, labelled and staged in the output row, reset the workspace before the next session.
- Clear the service station and re-run the Setup checklist before the next session.

After the episode: reset the workspace

This part is not recorded. It is just how you reset the table for the next episode.

- With recording off, move the serviced housings from the output zone back to the start zone for the next episode's config — back-left (Config L), front-centre (Config M), or back-right (Config R).
- Unscrew each sump, remove the new cartridge and set it back in the start zone as a fresh cartridge, then refit a practice cartridge and hand-tighten the sump, matching the initial setup state.
- Peel the date label off each cap and discard it, so each housing starts the next episode unlabelled.
- Empty the waste bin at the front-left and set any flagged housing aside for inspection; do not return it to the input zone.
- Return the wrench, the lubricant, the cloth, the label roll and the marker to the tool rest at the front-right.
- Confirm the table surface is clear of any objects other than the cradle, the tools on their rest, the waste bin, and the housings and cartridges in their designated zone.
- Go through the Setup checklist again before starting the next episode.

Warning: The table must be clear except for the cradle at the center, the tools on their rest at the front-right, the waste bin at the front-left, and the housings and cartridges in the start zone for the next episode's config.

Annotation subtasks

1. Move one housing from the input area to the service cradle
2. Unscrew the sump off the cap with the wrench and set it down
3. Lift the spent cartridge out and discard it in the waste bin
4. Pick one new cartridge and orient it gasket end up
5. Hand the new cartridge over to the right gripper above the service station (Config L and M)
6. Lower the new cartridge in square and centred on the standpipe
7. Lift, inspect, grease and seat the O-ring in its groove
8. Rescrew the sump hand-tight and snug it with the wrench
9. Wipe the sump and the cap dry
10. Write and apply the date label to the cap
11. Move the serviced housing to the output area
12. Home the arms